8/13 TeV NNLO theory predictions: content summary

Summary of the theory-prediction grid stored in `theory-data/output8/`, `theory-data/output13/` *and consolidated into* `theory-data/analysis/converted_data.json`. *Restricted to the ρ-related differential observables, inclusive cross sections, parameters and PDF variations; non-ρ differential distributions are omitted.*

Grid dimensions

- **Energies:** 8 TeV and 13 TeV.
- **Orders in the converted JSON:** `LO`, `NLO`, `NNLO_noVVF` (the full physical NNLO prediction). Raw bare `NNLO` ϵ-pole XMLs also exist on disk at both energies but are used only as an internal pole-cancellation check, not shipped in the converted dataset.
- **Top-quark mass grid:** 167.5, 171.5, 172.5, 173.5, 177.5 GeV, for every order and energy.
- **Scale variations:** 3 central scale choices (H_T, $H_T/2$, $H_T/4$), each with the standard 7-point (μ_R, μ_F) envelope, consolidated onto 13 distinct (μ_R, μ_F) points per mass/order. Nominal central scale is $H_T/2$.
- **Inclusive cross section:** ϵ-expansion coefficients $[c_0, c_{-1}, c_{-2}, c_{-3}, c_{-4}]$ from the STRIPPER dimensional-regularization output; c_0 (pb) is the physical cross section, $c_{-1} \dots c_{-4}$ are IR-pole coefficients kept as a numerical cancellation check.

ρ-related differential distributions

Label in JSON	Source key	Binning	Description
`rhocms8` / `rhocms13`	`rhocms`	1 binning, 4 bins	ρ with CMS-style selection, energy-tagged
`rhoatlas8`	`rhoatl` (binning 1)	8 bins	ρ with ATLAS-8 TeV-analysis binning
`rhoatlas13`	`rhoatl` (binning 2)	7 bins	ρ with ATLAS-13 TeV-analysis binning

`rhoatlas8` and `rhoatlas13` are stored at *both* theory energies (the "8"/"13" denotes the ATLAS binning scheme, not the collision energy); `rhocms` is energy-tagged instead. ρ is the top-mass-sensitive jet observable of Alioli, Fernandez, Fuster, Irles, Moch, Uwer, Vos, *A new observable to measure the top-quark mass at hadron colliders*, Eur. Phys. J. C 73 (2013) 2438, arXiv:1303.6415.

PDF sets

Eigenvector/error-set members are only fully populated at the physical central scale ($H_T/2$); off-diagonal scale points carry only the central (member 0) PDF for each family. LHAPDF IDs and member counts were cross-checked against LHAPDF's official `pdfsets.index` and match the member counts present in this repository's converted data; the literature references are the standard citations for these PDF groups.

Family (JSON key)	LHAPDF ID	Members stored	Role	Reference
`PDF4LHC21_40`	93100	41 (0–40: central + 40 eigenvectors)	PDF eigenvector set, all scale points	PDF4LHCWorki J. Phys. G 49 (202 arXiv:2203.05506
`CT18NNLO`	14000	59 (0–58: central + 29 eigenvector pairs)	PDF eigenvector set	Hou et al. (CT Phys. Rev. D 10 (2021), arXiv:1912
`CT18NNLO_as_0117`	14067	1	α_s down-variation of CT18NNLO	same
`CT18NNLO_as_0119`	14069	1	α_s up-variation of CT18NNLO	same
`MSHT20nnlo_as118`	27400	65 (0–64: central + 32 eigenvector pairs)	PDF eigenvector set	Bailey, Cridge, Lang, Martin, Th Phys. J. C 81 (arXiv:2012.04684
`MSHT20nnlo_as_smallrange`	27500	members 3, 4 only	α_s variation members (fine grid around 0.118)	same
`NNPDF31_nnlo_as_0118_hessian`	304400	101 (0–100: central + 100 Hessian eigenvectors)	PDF eigenvector set	NNPDF Coll. (Ba Eur. Phys. J. C 77 arXiv:1706.00428
`NNPDF31_nnlo_as_0117`	323100	1	α_s down-variation	same
`NNPDF31_nnlo_as_0119`	323500	1	α_s up-variation	same
`ABMP16als118_5_nnlo`	42780	30 (0–29: central + 29 eigenvectors)	PDF eigenvector set (nf=5)	Alekhin, Blümlei Placakyte, Phy D 96, 014011 arXiv:1701.05838
`ABMP16als117_5_nnlo`	42750	1	α_s down-variation	same
`ABMP16als119_5_nnlo`	42810	1	α_s up-variation	same